

PROJECT EXECUTIVE SUMMARY

Title: NeuroX-Fusion: Unified Foundation Model of Brain for Transformative Neuroscience

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Applying Institution/Organization: Brookhaven National Laboratory, The University of Texas at Austin, Seoul National University

Resource Name(s) and Number of Node Hours Requested: Aurora. 1,269,000 node-hours

Amount of Storage Requested: 600 TB

Executive Summary : Modern neuroscience is limited not by data scarcity but by the absence of a framework for integrating hundreds of terabytes scale, multimodal, multi-scale brain data into a single, computable object that captures the brain's dynamic architecture. *NeuroX-Fusion* will close this gap by training a 130 billion-parameter foundation model that unifies multimodal MRI and electrophysiology into one latent representation, enabling cross-modal decoding, mechanistic in-silico perturbations, and rapid hypothesis testing that were previously infeasible. Our scientific program is a staged campaign to construct this unified model. We will first develop two Large Brain Models (LBMs): **NeuroX-MRI**, built on our novel 4D Swin Transformer architecture to decode mesoscale brain dynamics from multimodal MRI data, and **NeuroX-Ephys**, a channel-equivariant model to decipher millisecond-scale neural syntax from heterogeneous electrophysiology recordings (ECoG, EEG, sEEG, and MEG). These two models will then be aligned through the shared semantic space of a Large Language Model (LLM), enabling the translation of latent brain states into explicit linguistic concepts. Our scientific program pursues three breakthrough objectives:

Decode the Unspoken Mind: Generate continuous, high-fidelity read-outs of internal affective and cognitive states and health from multimodal MRI and electrophysiology, enabling transformative applications, including early biomarkers for psychiatric disorders, novel brain therapeutics, and brain computer interfaces.

Bridge Brain Activity Across Scales and Modalities: Create the first unified model that fuses the high-resolution spatial data from MRI with the millisecond-scale temporal precision of electrophysiology. By aligning the disparate data streams, we will create a "digital twin" of brain function to probe the links between large-scale network dynamics and high-speed neural codes.

Advance Exascale AI for Science: Develop open-source novel AI architectures and data pipelines optimized for leadership-class systems. These innovations will create a transferable blueprint for data-centric AI in other complex DOE-relevant domains, such as climate science and materials discovery. To ensure broad adoption and reproducibility, all models and data pipelines will be integrated and disseminated through our established open-science platform, brainlife.io, democratizing access to these powerful tools for the global research community.

Converging these models requires training on over 400 TB scale of datasets, a process that necessitates the extreme I/O bandwidth and parallel file system capabilities unique to leadership-class systems like Aurora. Through our previous DOE ALCC award, we demonstrated excellent strong-scaling efficiency up to 512 nodes on Aurora, showing our computational readiness. We request 1,269,000 node-hours on Aurora over three years.

By creating a quantitative, computable model of the mind, NeuroX-Fusion will establish a new paradigm for neuroscience. An INCITE allocation is indispensable to converge this innovative integration of brain and AI, and ultimately, to transform the unspoken interior of the human mind into the domain of measurable, tractable science.

NeuroX-Fusion: Unified Foundation

S. Yoo, PhD

PROJECT NARRATIVE

1 SIGNIFICANCE OF RESEARCH

1.1. Grand Scientific Challenge: From Brain Mapping to Brain Decoding. Modern neuroscience is data-rich but knowledge-poor. We have over 400 terabytes of brain data from fMRI and EEG, but lack the ability to read its language. The fundamental barrier is the brain's staggering multi-scale complexity; a single thought unfolds across milliseconds (electrical spikes captured by Ephys), seconds (blood flow changes in fMRI), and is constrained by static anatomical wiring (dMRI). To truly understand the brain, we must unify these disparate data streams—yet no such scientific framework exists. The consequences are a global crisis in mental and neurological health, with conditions still diagnosed by subjective observation and an economic burden projected to reach \$16 trillion by 2030 [1]. This crisis demands a new class of scientific instrument: a unified, computational model of the brain. This proposal details the construction of that instrument, a foundation model designed to finally bridge the brain's multiple scales and translate its complex signals into measurable, tractable science.

1.2. Foundation Models for Predictive Models for Complex Systems. The fundamental gap between mapping and understanding the brain persists because the brain's complex, multi-scale dynamics render traditional methods ineffective [2, 3]. To address this, we propose to leverage foundation models, an AI paradigm that has become the essential tool for tackling such complexity. As demonstrated in domains from climate science to biology [4]-[5] and neuroscience [6, 7], large-scale models trained on massive heterogeneous data develop powerful emergent capabilities for prediction, decoding, and generation. This makes them uniquely suited to the challenges of modern neuroscience.

However, today's AI models for the brain remain fragmented. They are limited to single modalities and operate at capacities far below the regimes where advanced reasoning appears in language and vision [8]. Crucially, no existing model achieves the cross-modal, multi-scale integration needed to connect brain circuits, dynamics, and cognition altogether. Overcoming these limitations requires exascale computation. Our proposed NeuroX-Fusion model directly targets this gap, providing the first unified, 130 billion-parameter AI model designed to bridge disparate brain data with language representations at the exascale.

1.3. The Proposed Scientific Breakthrough: An Integrated, Multi-Scale Science Plan. Our project is built on a staged, three-pillar science plan to create a unified model of the mind.

Pillar I: NeuroX-MRI — Capturing the Dynamic Functional Connectome

The Scientific Gap: Systems neuroscience is dominated by the concept of the functional connectome, yet our analytical tools are fundamentally mismatched to its nature. We rely on methods that produce static, correlation-based "snapshots" of brain networks, which implicitly assume that connectivity is constant over the entire measurement period [3]. This approach fails to capture the brain's most critical property: its dynamic, moment-to-moment reconfiguration of neural circuits. This dynamic activity underpins the order and disorder of cognition, yet its neglect leaves us blind to the network instabilities that precede major neuropsychiatric disorders. Consequently, our understanding is limited to group-level biomarkers, falling short of the actionable, predictive insights that modern medicine demands.

Our Breakthrough Approach: We confront this limitation with NeuroX-MRI, a foundation model built on our novel Swin 4D fMRI Transformer architecture [9]. This approach is the first to learn

